

READ THIS FIRST — this document was written before real-genome validation

Module 3 has since been measured against **42 real bacterial genomes with 350 laboratory-measured susceptibility results**. On that test it scores **0.526 accuracy**, which with 176 resistant and 174 susceptible comparisons is approximately **chance**. No threshold changes this; the sweep peaks at 0.503.

Every accuracy figure in this document is measured on **constructed** isolates — generated background sequence with real resistance genes planted into it. Those figures are correct for constructed isolates and must not be presented as real-world performance.

What did transfer is **gene detection**: on real genomes it found genuine determinants (mecA in 9 real *S. aureus*, blaSHV in 5, plus blaNDM-1, blaVIM, blaCTX-M-15) at a 13.8% false-resistance rate, though it misses 73% of laboratory-confirmed resistance because 21 reference genes cannot cover real resistance. Full analysis: **reports/AUDIT_RESPONSE.md**.

TRIALSENSE • SEQUE 3.0 • QUESTION PREPARATION

Module 3 — Hard Questions and Prepared Answers

Every difficult question a judge could ask about the six features, split into knowledge questions and limitation questions. Answers are written to be spoken aloud, not read.

MARKER	MEANING	WHAT TO DO
SEVERE	Could damage credibility if fumbled	Memorise word for word
HARD	Needs a precise answer	Know the numbers
ROUTINE	Straightforward if you know the system	Read once

THE SINGLE MOST IMPORTANT RULE

If you do not know, say **“I do not know, and I would rather check than guess.”** One admitted gap makes every other answer more believable. One invented answer makes every other answer suspect.

01 Feature 1 — Runway

Projects resistance forward to the year the drug would launch.

HARD

What exactly are you fitting, and why that shape?

SAY

“An S-shaped curve, fitted by least squares after transforming the percentages so the problem becomes linear. We use an S-shape because resistance cannot go above 100%. Klebsiella carbapenem resistance rises about 2.5 points a year — a straight line extended to 2035 gives over 110%, which is not a number. The S-curve flattens as it approaches a ceiling, which is also what the real process does.”

Why they ask: to find out whether you chose the model or just used a default.

HARD

You excluded 2015. Why? That looks like dropping inconvenient data.

SAY

“It measures a different thing. 2015 covered only blood and spinal-fluid samples; 2016 onward covers all sample types. Joining them creates a false step — the MRSA figure reads 42.6% for 2015 and 28.4% for 2016, and both are correct for their own sample set. We still display the 2015 point, marked as excluded, rather than deleting it.”

Why they ask: excluding data is the classic way to make a trend look better. Showing the point rather than hiding it is the answer.

ROUTINE

How did you test the forecast?

SAY

“We fitted it using only data up to 2020, then asked it to predict 2021 to 2024 — four years it had never seen. 184 predictions across 46 series. Mean error 4.93 percentage points, and 85% landed within ten points.”

SEVERE

You have nine years of data and you are projecting eleven years forward. Justify that.

SAY

“You are right that it is a long extrapolation, and we do not present it as a measurement. It is drawn as a dashed line inside a widening uncertainty band precisely so it cannot be mistaken for data.”

“What we can defend is the near term, because we measured it — four years ahead, within five percentage points. Beyond that the honest statement is that the direction is reliable and the exact figure is not. For a go or no-go decision, direction is usually what matters.”

Why they ask: this is the strongest technical objection to Feature 1. Concede the point immediately, then narrow to what you measured.

HARD

Your sources disagree by fifteen points. Which one is right?

SAY

“We do not know, and neither does anyone else. ICMR, WHO and the national disease centre report different figures for the same organism and year, because they sample different hospitals and different specimen types. We show all of them and let the user choose, rather than picking one silently.”

“Disagreement between surveillance programmes is documented and normal. The same is true of the established gene-detection tools — two government tools run on identical data produce different gene lists.”

HARD

What breaks your projection?

SAY

“A new drug entering the market, a policy change on antibiotic use, or an outbreak of a different strain. The model assumes the trend continues, and it cannot detect any of those. It should be re-run when new surveillance data is published, not treated as a permanent answer.”

Does this work for every bacterium?

SAY

“No — seven, because that is what ICMR reports. Tuberculosis is not one of them, and for it we return ‘no data available’ rather than substituting a similar organism.”

02 Feature 2 – Rescue

Determines whether detected resistance can be worked around.

KNOWLEDGE QUESTIONS

SEVERE

Explain the difference between the two enzyme types – in your own words, no jargon.

SAY

“Both types are proteins the bacterium makes to destroy the antibiotic. They differ in how they are built. One type has a standard protein structure at its active site, and there are existing compounds that plug that site. The other uses a zinc atom to do the cutting, and those same compounds do not bind to it.”

“So the practical difference is: for the first type you add a second compound and the drug works again. For the second type there is nothing available to add. Same result on screen, opposite business decision.”

Why they ask: this is your headline claim. If you can only repeat the technical terms without explaining them, the claim collapses. **Practise this one aloud.**

HARD

Who decided these relationships? Did you?

SAY

“No. They come from two published databases – CARD, maintained by a university research group, and NCBI's reference gene catalogue. Every relationship in our project has the source recorded with a link. We wrote the first version from textbook references and then checked all 21 genes against both databases.”

SEVERE

What if one of your 21 entries is wrong? Everything downstream would be wrong.

SAY

“Correct, and that is why we cross-checked all 21 against two independent databases rather than trusting our own notes. The check found genuine differences, and we recorded them rather than silently picking one.”

“It also found one error in our own labelling — a resistance marker was labelled with a position number that belongs to a different organism. We fixed it. That is the process working.”

Why they ask: a lookup table is only as good as its contents. Naming a mistake you found is the strongest possible answer.

HARD

You said you cannot detect single-letter changes. Isn't that a large share of real resistance?

SAY

“Yes, and it is a genuine limit. Two of our 21 markers work that way. Our method counts short letter patterns across a whole region, and a single letter changing does not shift those counts measurably.”

“So for those two we report that the region is present, not that the resistant version is present. We say that explicitly rather than implying we detected something we did not. Resolving it needs a different technique that reads individual positions.”

Why they ask: it is a real technical gap. Knowing exactly which two genes and exactly why is what separates understanding from recitation.

ROUTINE

What happens with a gene not in your list?

SAY

“No mechanism assessment — we return nothing rather than guessing. Feature 4 may still flag the region as unidentified foreign DNA, which is a weaker but honest statement.”

03 Feature 3 — Portfolio

Screens every candidate against every target organism.

KNOWLEDGE QUESTIONS

HARD

How do you rank the candidates, and why that rule?

SAY

“By how many organisms the candidate remains usable against, with lower average predicted resistance as the tie-break.”

“I will be straight with you — that is our judgement, not a validated rule. No evidence says breadth is the correct primary criterion. It is a starting point a user would adjust to their own priorities.”

Why they ask: ranking rules often hide unexamined assumptions. Conceding it is a choice is stronger than defending it as science.

LIMITATION QUESTIONS

SEVERE

This feature has no data or model of its own. So what have you actually validated?

SAY

“That is exactly right, and it is why we validated it differently. Every prediction in the grid is compared against the real ICMR national figure for the same organism and antibiotic group — 49 comparisons.”

“We judge each one against its expected direction rather than expecting the numbers to match, because ours describes one sample and ICMR describes a national population. A sample chosen because it is resistant should read higher than a national average.”

Why they ask: orchestration features are usually unvalidated. Yours is not — but only because we built that check specifically.

HARD

Your cross-check flagged sixteen of forty-nine. That is a third. Explain.

SAY

“Most of those are expected. Our samples are built to carry resistance genes, so they should read above a national average that includes ordinary samples. The check separates differences that go the direction we predicted from those that do not.”

“The ones that genuinely concerned us we investigated. One showed our grouping was too coarse in one place, and we fixed it — we now report individual drugs rather than whole families where the reference data supports it.”

ROUTINE

Does this account for cost or manufacturability?

SAY

“No. Resistance only. A real portfolio decision also weighs manufacturing cost, regulatory position and market size, and none of that is in here.”

04 Feature 4 — Unknown

Flags DNA that looks acquired but matches nothing known.

KNOWLEDGE QUESTIONS

HARD

Why count four-letter patterns? Why not three, or six?

SAY

“Four is the conventional choice for this kind of comparison, and there is a practical reason. Four letters gives 256 possible combinations. In a 900-letter window that is roughly three or four counts per combination, which is enough to be stable.”

“Five letters gives 1,024 combinations, so most of them appear zero or once — the measurement becomes mostly noise. We tested this and five performed noticeably worse.”

ROUTINE

Why do you use two thresholds instead of one?

SAY

“Because the relative measure becomes unstable when few windows remain — a small spread inflates it. We require a window to be both statistically unusual and absolutely different. On our test set the absolute measure separated cleanly where the relative one did not.”

SEVERE

You tested it by planting sequences you wrote yourself. That is circular.

SAY

“It is a fair criticism and I will not argue with it. We have never tested it against a genuinely novel resistance gene, because we do not have one — by definition.”

“What the test does establish is that the detector responds to foreign composition rather than to anything specific about our reference set, and that it stays silent on all twelve unmodified samples. The stronger test would be to withhold a real gene from the reference set and see if it is flagged. That is the next thing we would do.”

Why they ask: self-generated test data is a real weakness. Naming the better experiment shows you understand why.

HARD

If it fires, what does the user actually do? It sounds like it just creates worry.

SAY

“It reports the position — start and end. The action is to sequence that region properly and identify it, which is a targeted piece of laboratory work rather than an open-ended search.”

“The alternative is worse. Every tool in this category reports only what is in its database, so silence from one of them is indistinguishable from confirmation that nothing is there. Those are very different statements, and a company proceeding on the wrong one is the expensive case.”

HARD

What is your false alarm rate on real genomes?

SAY

“We do not know. Zero across our twelve samples, but those are constructed, and a real genome contains far more naturally occurring foreign regions that have nothing to do with resistance. I would expect the rate to be higher on real data and we have not measured it.”

Why they ask: a tempting question to bluff. Do not. The honest answer is short and costs nothing.

05 Feature 5 – Value

Converts a finding into a rupee figure. The most attackable feature.

BEFORE ANY QUESTION IS ASKED

Say the disclaimer **first**, unprompted: “This is a calculator, not a measurement.” A judge who hears it from you cannot use it against you.

KNOWLEDGE QUESTIONS

HARD

Where do your stage costs come from?

SAY

“They are order-of-magnitude figures for anti-infective development, supplied as starting values. They are not from a specific study, and I would not defend any individual number. Every one is a slider the user replaces with their own figures.”

LIMITATION QUESTIONS

SEVERE

This number is made up.

SAY

“You are right not to trust it, and that is why it is a slider rather than a headline. What are your numbers? Let us put them in.”

“The claim we would actually defend does not depend on any of them: running this check costs effectively nothing, and not running it costs at minimum one failed development stage. That gap holds whatever figures you use.”

Why this works: agreeing instantly and offering to change the inputs live converts an attack into a demonstration. Do not defend the number.

HARD

Why assume the problem surfaces at that particular stage?

SAY

“Because that is the first stage where the drug is tested against patients who actually have the infection. Earlier stages test safety, not whether it works on the target. It is a reasoned assumption and it is also a dropdown the user can change.”

HARD

Expected value assumes the company is indifferent to risk. Pharmaceutical companies are not.

SAY

“That is a good point and we have not modelled it. A smaller company facing a single large loss behaves very differently from one running twenty programmes. Our figure treats them the same, which understates the value for the smaller company — which is the user we are actually aiming at.”

Why they ask: this comes from someone with a finance background. Recognising it as correct earns more than deflecting.

06 Feature 6 — Spread

Reports how quickly a resistance gene moves through a population.

KNOWLEDGE QUESTIONS

ROUTINE

In your own words, what is the difference between the two kinds of gene?

SAY

“Some resistance genes sit on a small separate loop of DNA that a bacterium can pass directly to another bacterium, including one of a completely different species. Others are fixed inside the main chromosome and only pass to that bacterium's own descendants. The first kind spreads far faster.”

LIMITATION QUESTIONS

SEVERE

Where does the number 2.5 come from?

SAY

“It is a reasoned estimate of relative magnitude, not a measured value, and the interface says so on the same screen.”

“What is defensible is the ordering — transferable genes demonstrably spread faster than fixed ones. One well-documented example went from first description in 2015 to reports on multiple continents within about eighteen months, which no fixed mutation does. What we cannot defend is that the factor is exactly 2.5 rather than 2 or 3. Deriving it from observed spread rates is on our list.”

Why they ask: an unexplained constant invites exactly this. Conceding the specific number while defending the ordering is the correct split.

HARD

The eighteen-month figure — is that your data or someone else's?

SAY

“Someone else's. It is a documented published observation, not something we measured. We have measured nothing about spread rates ourselves.”

07 Questions about the whole module

These are the ones most likely to be asked, and the ones that decide the round.

SEVERE

You have no healthcare background. Why should I believe any of this?

SAY

“Because we did not decide any of the biology. Every gene sequence comes from NCBI's reference database with a catalogue number you can look up. Every resistance relationship comes from two published databases. Every resistance figure comes from ICMR's published reports.”

“What we built is the engineering — the pipeline, the validation, and the decision logic on top of settled science. That is what engineers do.”

SEVERE

What is the single weakest part of Module 3?

SAY

“The training data. The gene sequences are real, but the DNA around them is constructed and we assign which genes each sample carries — so the labels are ours, not laboratory-confirmed.”

“That is why the model on its own has a missed-detection rate above the standard applied to commercial testing equipment. We do not ship it on its own; the gene search overrides it. The fix is complete public genomes that come with laboratory results attached.”

Why they ask: to see whether you know. Having a specific answer ready is worth more than the weakness costs you.

SEVERE

If I handed you a real bacterial genome right now, would this work?

SAY

“The pipeline would run — it accepts a standard sequence file and the same code path executes. The gene search should perform well, because it compares against real published sequences.”

“The statistical model would perform worse than our figures suggest, because a real genome is millions of letters with noise and incomplete regions, and ours are clean and short. I would not claim otherwise.”

Why they ask: the most direct test of whether you understand your own limits. A confident “yes, it works” here would be a lie.

HARD

Which of the six features would you delete?

SAY

“The cost calculator, if forced. It carries no data of its own and it attracts the most scepticism. We keep it because a scientific finding that cannot be stated in rupees does not reach the person who approves the budget — but it is the one that rests on the least evidence.”

HARD

Free government tools already do this. Why build another?

SAY

“They do the gene search better than we do, and we use their data. They are built for hospital staff treating a patient today.”

“They do not project to a launch year, they do not say whether the resistance can be worked around, and they do not compare a portfolio of candidates. Those three questions decide whether a company funds a programme, and that is the gap we built for.”

HARD

What did you get wrong during development?

SAY

“Several things, and we kept a record. The unknown-element detector originally scored clean samples higher than deliberately modified ones — it was rediscovering genes we had already found. We rebuilt it three times.”

“We also reported a drug group we do not check as having zero resistance rather than as unchecked, and we had a position number that belongs to a different organism. Eight defects in total, all found by our own testing, all documented.”

Why this is a good question to get: a validation process that never finds anything is not evidence of correctness. This is your strongest answer — do not shorten it.

SEVERE

Would you bet two hundred crore on this output?

SAY

“No, and it is not designed for that. It is a screening tool — it tells a team which questions to ask a laboratory first, not what to conclude.”

“What I would say is that spending two hundred crore *without* running this check is worse, because the check costs effectively nothing and takes seconds.”

Why they ask: a trap. Answering “yes” overclaims immediately. The correct answer reframes what the tool is for.

08 Questions built on a wrong assumption

Correct the premise politely, then answer. Do not accept a false framing.

THEY SAY	THE ERROR	YOUR CORRECTION
“So your AI predicts all six of these things.”	There is one model, not six	“One of the six uses a trained model. The rest are similarity matching and published lookups — deliberately, because encoded science beats a guess where the science is settled.”
“Your accuracy is 99%.”	That was the old figure on constructed sequences	“It was, on constructed sequences. On real ones it is 0.84 on the hardest test. We replaced the data and kept the lower number.”
“You are using real patient genomes.”	Real genes, constructed scaffolds	“Real gene sequences with catalogue numbers, placed into constructed surrounding DNA. No patient data at all.”
“So this tells doctors which antibiotic to give.”	Wrong user entirely	“No — it is for R&D teams deciding what to develop. It is not a diagnostic and nothing here should be used to treat anyone.”
“You have validated this against lab results.”	Never claimed, and not true	“We have not. We validated against published national statistics and against labels we assigned ourselves. Laboratory confirmation would need the public genome database, which is our next step.”

WHY THIS SECTION MATTERS MOST

A judge restating your work incorrectly, in good faith, is the most likely way an overclaim enters the record. If you nod along, that becomes what you claimed. Correcting a compliment is uncomfortable and it is the right move every time.

09 What to practise, in order

PRIORITY	QUESTION	WHY FIRST
1	“Explain the two enzyme types in your own words”	Your headline claim collapses without it
2	“Why should I believe you with no healthcare background”	Most likely question in the room
3	“What is the weakest part”	Sets up your strongest answer
4	“This cost number is made up”	Only works if the reply is instant
5	“Would you bet two hundred crore on it”	A trap that catches unprepared teams
6	“What did you get wrong”	Your best material — do not improvise it

METHOD

Have a teammate read the question aloud and answer without looking. If you pause to think for more than two seconds, you have not learned it yet. Six questions, three rounds each.

All figures quoted in these answers are reproducible from `models/metrics.json` and `models/validation.json` by running `python train.py` and `python validate_module3.py`.

TrialSense is a research screening prototype. It is not a diagnostic tool and not a clinical decision-support system.